BABA MASTNATH UNIVERSITY

Faculty of Engineering & Technology

B.Tech 3rd Semester – Sessional-1

Subject: Computer Organization & Architecture (COA)

Time: 1:30 Hrs | MM: 30

Q1. Draw a block diagram and explain in detail the digital computer system architecture. (10 Marks)

Block Diagram of Digital Computer System

```
            +------------------+
            |   Input Unit     |
            +------------------+
                     |
                     v
        +---------------------------------+
        |        Central Processing Unit |
        |                                 |
        |  +-----------+  +-----------+ |
        |  | Control   |  |    ALU    | |
        |  |   Unit    |  |           | |
        |  +-----------+  +-----------+ |
        |            |        |           |
        |            +---- Registers -----+
        +---------------------------------+
                     |
                     v
            +------------------+
            |  Main Memory     |
            | (RAM / ROM)      |
            +------------------+
                     |
                     v
            +------------------+
            |  Output Unit     |
            +------------------+
```

Explanation

A digital computer system is an electronic machine that processes data using binary signals. The architecture of a digital computer defines how different functional units are interconnected and how they operate together to perform computations.

The main components of a digital computer system are explained below.

1. Input Unit

The input unit is responsible for receiving data and instructions from the user. It converts the input data into binary form so that it can be understood by the computer system. This unit acts as an interface between the user and the computer. Common input devices include keyboard, mouse, scanner, and microphone.

2. Central Processing Unit (CPU)

The CPU is the core component of the digital computer system and is often called the brain of the computer. It performs all arithmetic, logical, and control operations.

(a) Control Unit (CU)

The control unit controls and coordinates the activities of all parts of the computer. It fetches instructions from memory, decodes them, and generates control signals. These signals direct the ALU, memory, and input/output devices to perform the required operations in a proper sequence.

(b) Arithmetic Logic Unit (ALU)

The ALU performs all arithmetic operations such as addition, subtraction, multiplication, and division. It also performs logical operations like AND, OR, NOT, and comparison operations.

(c) Registers

Registers are small, high-speed storage units located inside the CPU. They temporarily store data, instructions, and intermediate results during processing, thereby improving system speed.

3. Main Memory

Main memory stores data, instructions, and intermediate results required during execution. It includes RAM, which is volatile, and ROM, which is non-volatile. Memory plays a crucial role in the overall performance of the system.

4. Output Unit

The output unit presents the processed information to the user in a human-readable form. Examples include monitor, printer, and speakers.

Q2. Explain what you understand by data representation and various number systems. (10 Marks)

Data Representation

Data representation refers to the way data such as numbers, characters, and symbols are represented inside a computer system. Since computers work on binary signals, all types of data are converted into binary form.

Different forms of data representation include:

Numeric data representation

Character representation

Boolean data representation

Proper data representation is necessary for efficient processing and storage of information.

Number Systems

A number system is a method of representing numbers using a set of digits and a base (radix). The commonly used number systems in computers are explained below.

1. Decimal Number System

The decimal number system uses base 10 and digits from 0 to 9. It is commonly used by humans in daily life.

Example: (345)■■

2. Binary Number System

The binary number system uses base 2 and digits 0 and 1. It is the fundamental number system used by computers.

Example: (10101)■

3. Octal Number System

The octal number system uses base 8 and digits from 0 to 7. It provides a compact representation of binary numbers.

Example: (745)■

4. Hexadecimal Number System

The hexadecimal number system uses base 16 and digits 0–9 and A–F. It is widely used in memory addressing and machine-level programming.

Example: (2AF)■■

Q3. Explain with the help of block diagram the common bus system. (10 Marks)

Block Diagram of Common Bus System

```
Register A ----|
Register B ----|
Register C ----|
```

```
                   |---- Common Bus ----> ALU
Register D ----|
Register E ----|

        ^
        |
   Control Unit
```

Explanation

A common bus system is a communication mechanism used to transfer data between various components of a computer system using a single shared data path known as a bus.

Instead of providing separate connections between every register, a common bus reduces the number of interconnections and simplifies hardware design.

The control unit generates control signals that determine:

Which register will place data on the bus

Which register will receive data from the bus

At any time, only one register is allowed to transmit data on the bus to avoid data conflicts.

Advantages

Reduces hardware complexity

Cost-effective design

Easy to control and implement

Disadvantages

Only one transfer can occur at a time

Bus contention may cause delays

Q4. Explain with the help of block diagram the time and control unit of a basic computer. (10 Marks)

Block Diagram of Time and Control Unit

```
       +-------------------+
       |   Instruction     |
       |     Register      |
       +-------------------+
                 |
                 v
       +-------------------+
       |   Decoder         |
       +-------------------+
                 |
                 v
       +-------------------+
       | Control Logic     |
       +-------------------+
                 |
                 v
+-------------------------------------+
| Timing Signals (T0, T1, T2, ...)    |
+-------------------------------------+
                 |
                 v
      Control Signals to CPU Units
```

Explanation

The time and control unit is a part of the control unit of a basic computer. It is responsible for generating timing and control signals required for the execution of instructions.

The instruction stored in the instruction register is decoded by the decoder. Based on the decoded instruction and timing signals, the control logic generates appropriate control signals.

Timing signals such as T■, T■, T■, etc., ensure that micro-operations are executed in the correct sequence. This coordination ensures smooth functioning of the CPU and proper execution of instructions.

Conclusion

The time and control unit plays a crucial role in synchronizing and controlling all operations of the basic computer system.